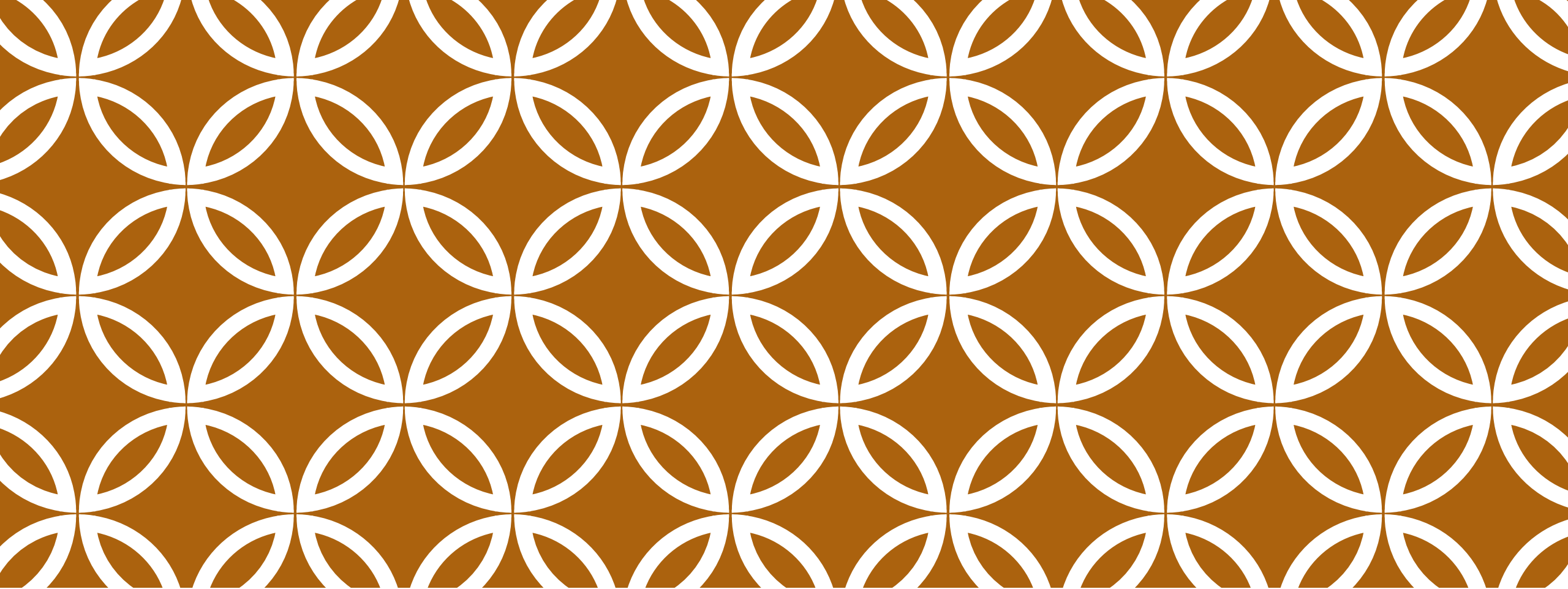




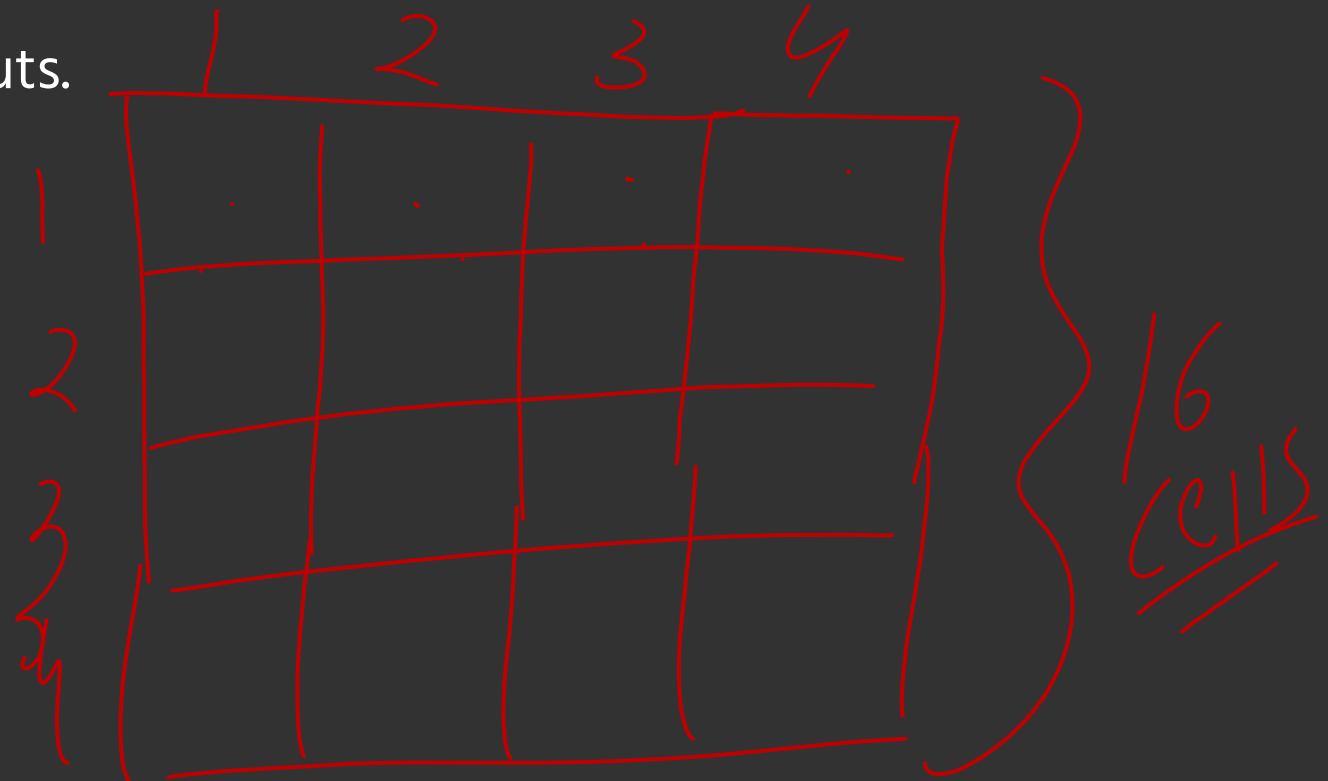
MINIMIZE BOOLEAN EXPRESSIONS - 3

Unit – 2: Digital Electronics

4-VARIABLE K-MAP

0 → 15 ⇒ Hex

- ✓ A 4-variable expression can have $2^4 = 16$ possible ways of input representation.
- ✓ Thus, a 4-variable K-Map will have 16 adjacent cells representing each of the 16 possible combinations of inputs.



Representation using minterms (SOP form)

minterms

<u>AB</u> \ <u>CD</u>	<u>$\bar{C}\bar{D}$</u>	<u>$\bar{C}D$</u>	<u>$C\bar{D}$</u>	<u>CD</u>
<u>$\bar{A}\bar{B}$</u>	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
<u>$\bar{A}B$</u>	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BC\bar{D}$	$\bar{A}BCD$
<u>AB</u>	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABC\bar{D}$	$ABCD$
<u>$A\bar{B}$</u>	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$A\bar{B}C\bar{D}$	$A\bar{B}CD$

<u>AB</u> \ <u>CD</u>	<u>00</u>	<u>01</u>	<u>11</u>	<u>10</u>
<u>00</u>	<u>m_0</u>	<u>m_1</u>	<u>m_3</u>	<u>m_2</u>
<u>01</u>	<u>m_4</u>	<u>m_5</u>	<u>m_7</u>	<u>m_6</u>
<u>11</u>	<u>m_{12}</u>	<u>m_{13}</u>	<u>m_{15}</u>	<u>m_{14}</u>
<u>10</u>	<u>m_8</u>	<u>m_9</u>	<u>m_{11}</u>	<u>m_{10}</u>

0 $\rightarrow$ x'
1 $\rightarrow$ x

Representation using minterms (POS form)

Minterms

<u>AB</u> \ <u>CD</u>	<u>C+D</u>	<u>C+D̄</u>	<u>C̄+D</u>	<u>C̄+D̄</u>
<u>A+B</u>	A+B+C+D	A+B+C+D̄	A+B+C̄+D	A+B+C̄+D̄
A+B̄	A+B̄+C+D	A+B̄+C+D̄	A+B̄+C̄+D	A+B̄+C̄+D̄
Ā+B	Ā+B+C+D	Ā+B+C+D̄	Ā+B+C̄+D	Ā+B+C̄+D̄
Ā+B̄	Ā+B̄+C+D	Ā+B̄+C+D̄	Ā+B̄+C̄+D	Ā+B̄+C̄+D̄

AB \ CD	<u>00</u>	01	11	10
<u>00</u>	M ₀ 0	M ₁ 1	M ₃ 3	M ₂ 2
01	M ₄ 4	M ₅ 5	M ₇ 7	M ₆ 6
11	M ₁₂ 12	M ₁₃ 13	M ₁₅ 15	M ₁₄ 14
10	M ₈ 8	M ₉ 9	M ₁₁ 11	M ₁₀ 10

0 → X
1 → X

Example: Reduce the function $F(A,B,C,D) = \Sigma(0,1,2,3,12,13,15,14)$ using K-Map

- ◆ Here the 3 variables as in the function so we draw 3 variable k-map and fill ones in the given minterms: m_0 to m_3 and m_{12} to m_{15} .

CD	00	01	11	10
AB 00	1	1	1	1
AB 01	0	0	0	0
AB 11	1	1	1	1
AB 10	0	0	0	0

◆ First Quad =

$$A'B' \leftarrow F_1$$

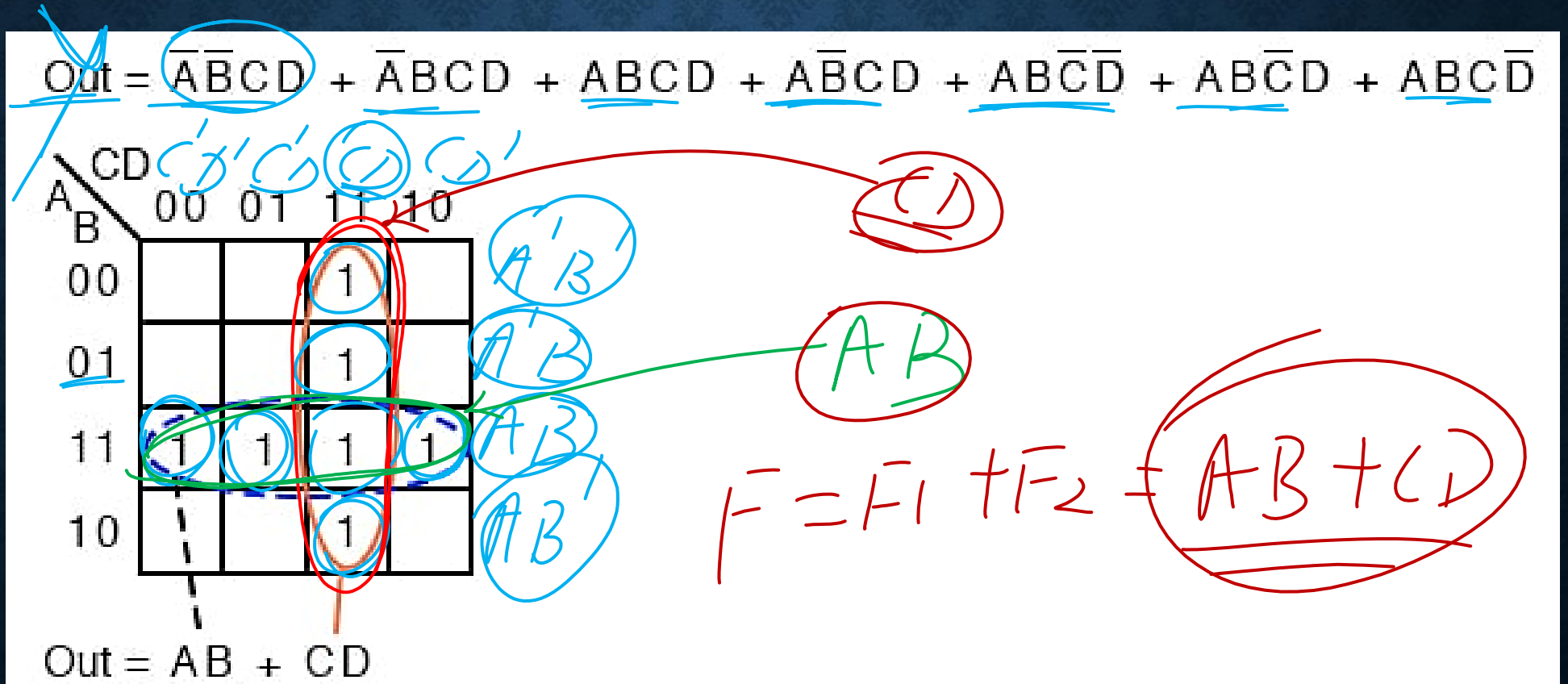
◆ Second Quad =

$$AB \leftarrow F_2$$

$$F = F_1 + F_2$$

$$= A'B' + AB$$

EXAMPLE: REDUCE THE FUNCTION USING K-MAP



AB \ CD		CD			
		[00]	[01]	[11]	[10]
AB	[00]	1 0	0 1	1 3	1 2
	[01]	1 4	0 5	1 7	1 6
	[11]	1 12	0 13	0 15	0 14
	[10]	1 8	0 9	0 11	0 10

AB \ CD		CD			
		[00]	[01]	[11]	[10]
AB	[00]	1 0	0 1	1 3	1 2
	[01]	1 4	0 5	1 7	1 6
	[11]	1 12	0 13	0 15	0 14
	[10]	1 8	0 9	0 11	0 10

• The 1st quad = $C'D'$

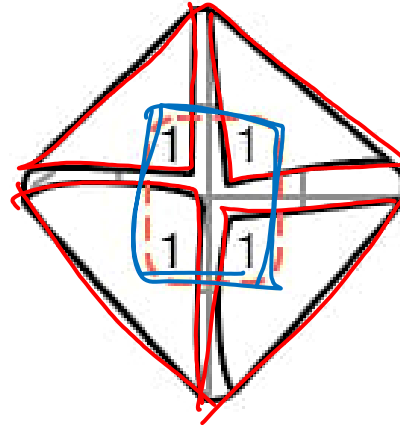
• The 2nd quad = $A'C$
 $\equiv A'C + C'D'$

$$\begin{aligned}
 (A'CD) + (A'CD') &= A'C(\underline{D} + \underline{D'}) \\
 &= A'C(1) \\
 &= A'C
 \end{aligned}$$

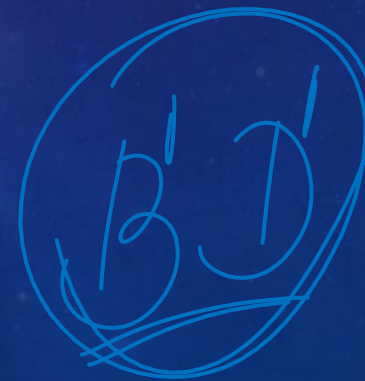
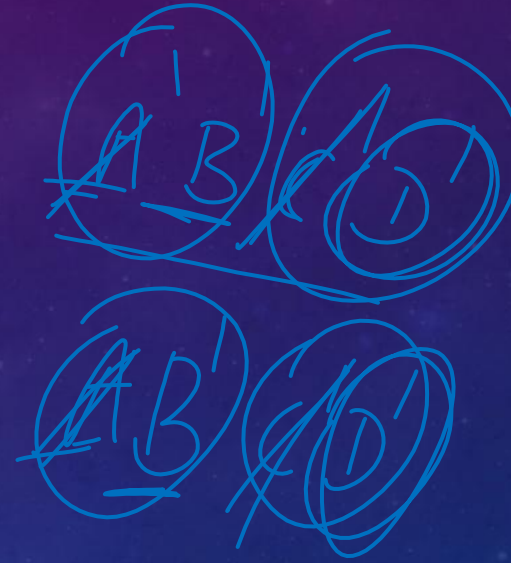
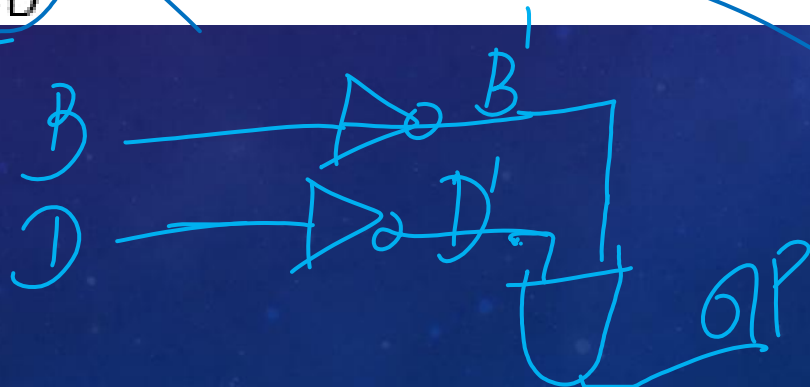
SIMPLIFY THE SOP FUNCTION USING K-MAP

$$\text{Out} = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + A\overline{B}\overline{C}\overline{D} + A\overline{B}C\overline{D}$$

A \ B	CD			
	00	01	11	10
00	1			1
01				
11				
10	1			1



$$\text{Out} = \overline{B}\overline{D}$$



$$\text{Out} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD$$

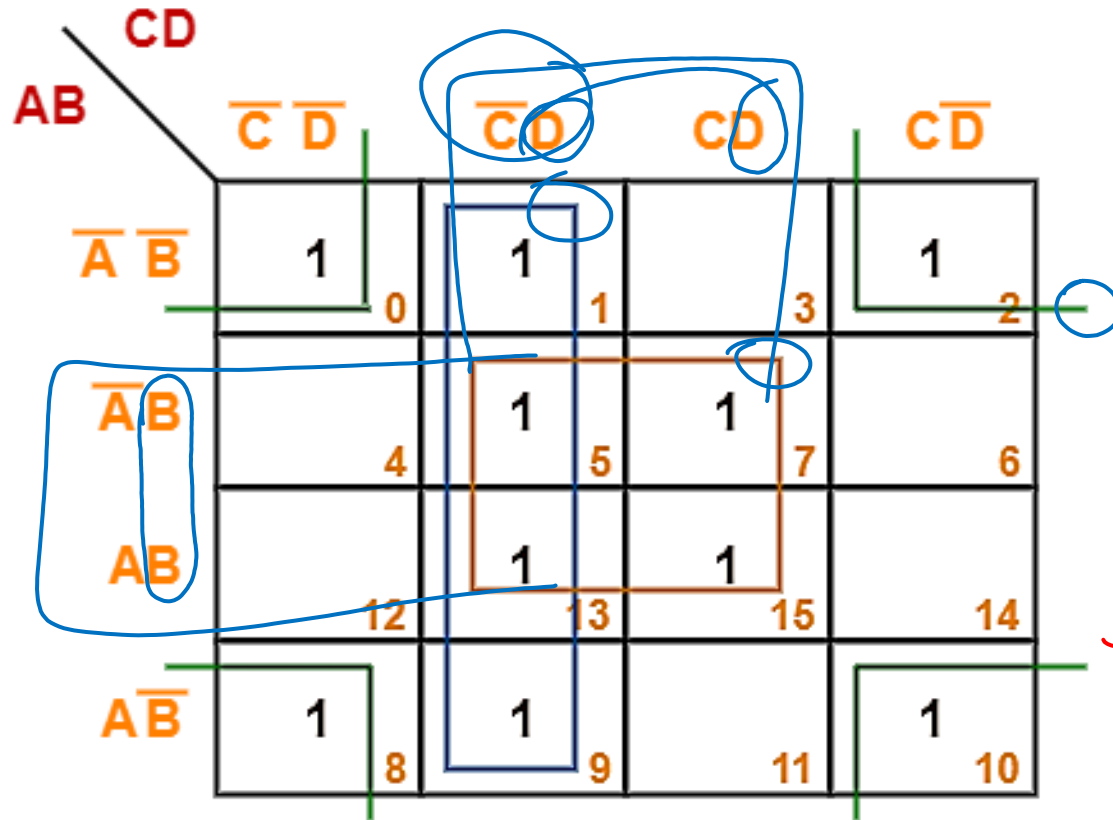
A \ B		CD			
		00	01	11	10
00	1	1	1	1	1
01					
11					
10	1	1	1	1	1

~~Out = B~~

B

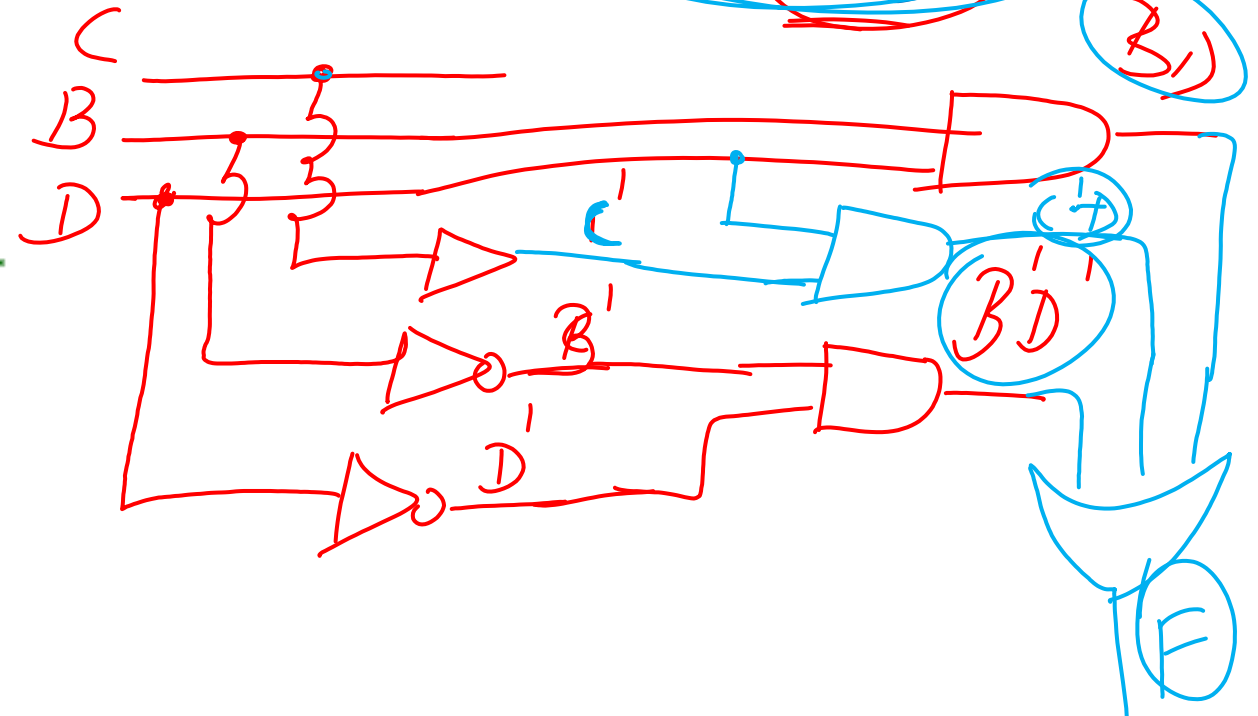


Minimize the following Boolean function
 $F(A, B, C, D) = \Sigma m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$



O/P

$$F = BD' + B'D + CD$$



- ✓ ◇ Reduce the Boolean function: $F(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$
- ✓ ◇ Minimize the Boolean function: $f(A, B, C, D) = \sum m(2, 3, 6, 7, 8, 10, 13, 15)$
- ✓ ◇ Simplify the Boolean function $F(w, x, y, z) = \sum (4, 5, 6, 7, 12, 13, 14, 15)$

H/E

CD \ AB		CD			
		[00]	[01]	[11]	[10]
[00]	0	0	1	1	0
[01]	4	1	0	0	1
[11]	12	1	0	0	1
[10]	8	0	1	1	0

H/E

4-var
Karnaugh
Book XXX

MAXTERM (POS) SOLUTION OF K MAP

Handwritten notes: "maxterms" with an arrow pointing to a circled "0's", and "0's" circled in red.

Example : $Y = (A + B + \overline{C}) + (A + \overline{B} + \overline{C}) + (\overline{A} + \overline{B} + C) + (\overline{A} + \overline{B} + \overline{C})$

Handwritten red annotations below the expression: "0+0+1" under the first term, and "0 1 1" under the second term. Further "1 1 0 1 1 1" are written below the remaining terms.

$\rightarrow A$
 $\rightarrow A'$

A \ BC	BC			
	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Handwritten annotations on the K-map: Yellow circles around the 0s in cells (0,1), (0,3), (1,7), and (1,6). Green rectangles group the 0s in columns BC=01 and BC=11. Red checkmarks are placed below the grouped 0s. Red scribbles are over the 0s in cells (0,0), (0,2), (1,4), and (1,5). To the right, a red circle contains a diagram showing a 0 and a 1 with arrows pointing to a circled 'X'.

Simplified Expression : $Y = (A + \overline{C})(\overline{A} + \overline{B})$

$Y = (A + C') \cdot (A' + B')$

Handwritten notes: "Product of sum" written vertically to the right of the expression. Arrows point from the terms in the expression to the words "Product" and "sum".

SIMPLIFY THE POS FUNCTION $F(A,B,C) = \prod (0,1,6,7)$

A \ BC	$\overline{B}\overline{C}$	$\overline{B}C$	$B\overline{C}$	BC
$\overline{A}$	0	1	3	2
A	4	5	7	6

$$F = (A' + B') \cdot (A + B)$$

$$\overline{F} = \overline{A}\overline{B} + \overline{A}B$$

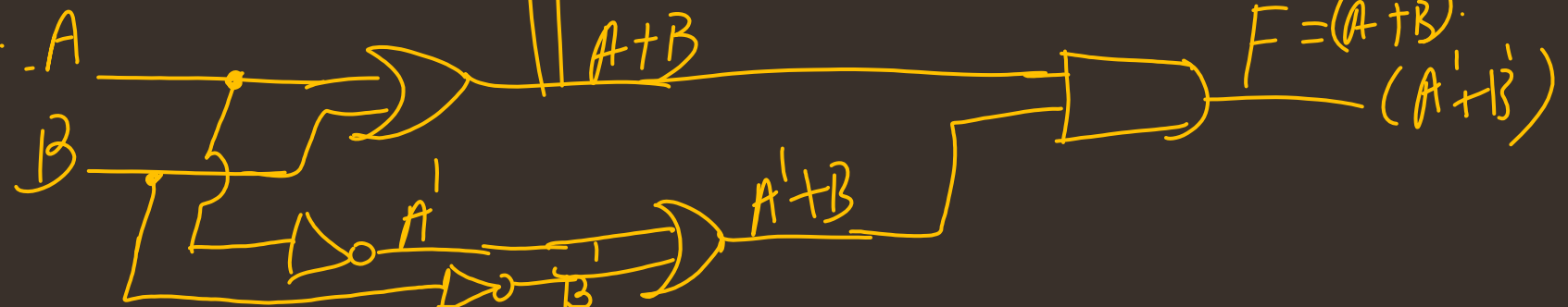
$$\overline{\overline{F}} = \overline{\overline{A}\overline{B} + \overline{A}B}$$

$$F = (\overline{\overline{A}\overline{B}}) + (\overline{\overline{A}B})$$

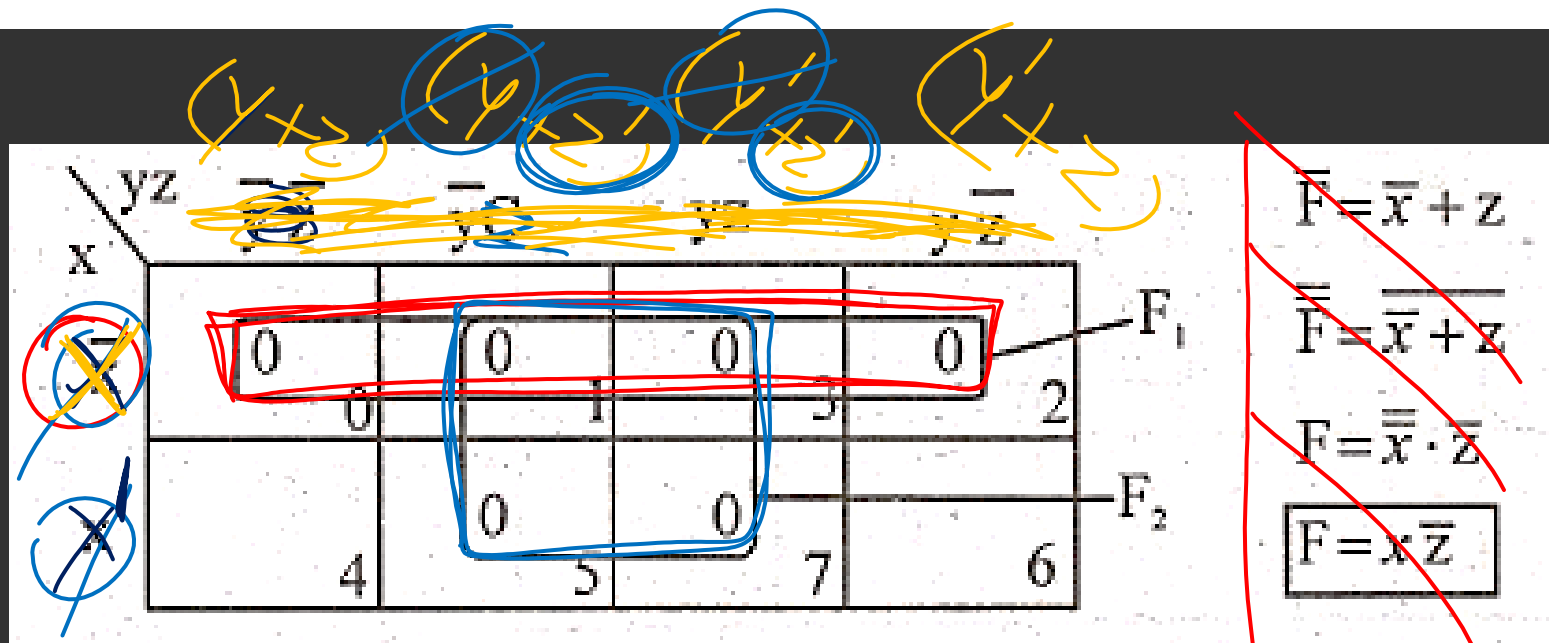
$$F = (\overline{\overline{A}\overline{B}}) + \overline{\overline{A}B}$$

$$F = (A + B) + (A + \overline{B})$$

Logic Diagram



SIMPLIFY THE POS FUNCTION $F(X,Y,Z) = \Pi(0,1,2,3,5,7)$



~~$$\bar{F} = \bar{x} + z$$~~

~~$$\bar{F} = \bar{x} + z$$~~

~~$$F = \bar{x} \cdot \bar{z}$$~~

~~$F = x \bar{z}$~~

[illegible]

$$F(A,B,C,D) = \Pi(3,5,7,8,10,11,12,13)$$

Handwritten notes on the left side of the K-map:

- $A+B$ (circled)
- $A+B'$ (circled)
- $A'B'$ (circled)
- $A+B'$ (circled)
- $A'B$ (circled)

AB \ CD	00	01	11	10
00	1 0	1 1	0 3	1 2
01	1 4	0 5	0 7	1 6
11	0 12	0 13	1 15	1 14
10	0 8	1 9	0 11	0 10

Handwritten groupings on the K-map:

- Red box around cells (0,3), (1,7), (2,11), (3,10) with label $A+B$.
- Green box around cells (1,5), (2,13), (3,9) with label $A+B'$.
- Blue box around cells (2,12), (3,8) with label $A'B'$.
- Orange box around cells (2,11), (3,10) with label $A'B$.

$$f_1 = A + C' + D'$$

$$f_2 = B + C + D'$$

$$f_3 = A + B + D$$

$$f_4 = A' + C + D$$

$$F = f_1 \cdot f_2 \cdot f_3 \cdot f_4$$

$$\text{Answer: } (C + D' + B').(C' + D' + A).(A' + C + D).(A' + B + C')$$

$\Sigma m(0,1,2,5,8,9,10) \& \pi(3,4,6,7,11,12,13,14,15)$

AB \ CD	$\overline{C}\overline{D}$	$\overline{C}D$	$C\overline{D}$	CD
$\overline{A}\overline{B}$	1	1	0	1
$\overline{A}B$	0	1	0	0
$A\overline{B}$	0	0	0	0
AB	1	1	0	1

F_1 points to the cell $(\overline{A}\overline{B}, CD)$
 F_2 points to the cell $(A\overline{B}, \overline{C}\overline{D})$
 F_3 points to the cell $(\overline{A}B, \overline{C}\overline{D})$

$$F = F_1 + F_2 + F_3$$

$$F_1 = (\overline{C} + \overline{D})$$

$$F_2 = (\overline{A} + \overline{B})$$

$$F_3 = (\overline{B} + D)$$

$$F = (\overline{C} + \overline{D})(\overline{A} + \overline{B})(\overline{B} + D)$$

2

AB \ CD	CD			
	00	01	11	10
00				
	0	1	3	2
01	0			0
	4	5	7	6
11			0	0
	12	13	15	14
10			0	
	8	9	11	10

f_1

f_2

f_3

Figure 3 - POS K-Map

- Answer: $f(A,B,C,D) = (A+B^-+D)(A^-+C^-+D^-)(A^-+B^-+C^-)$



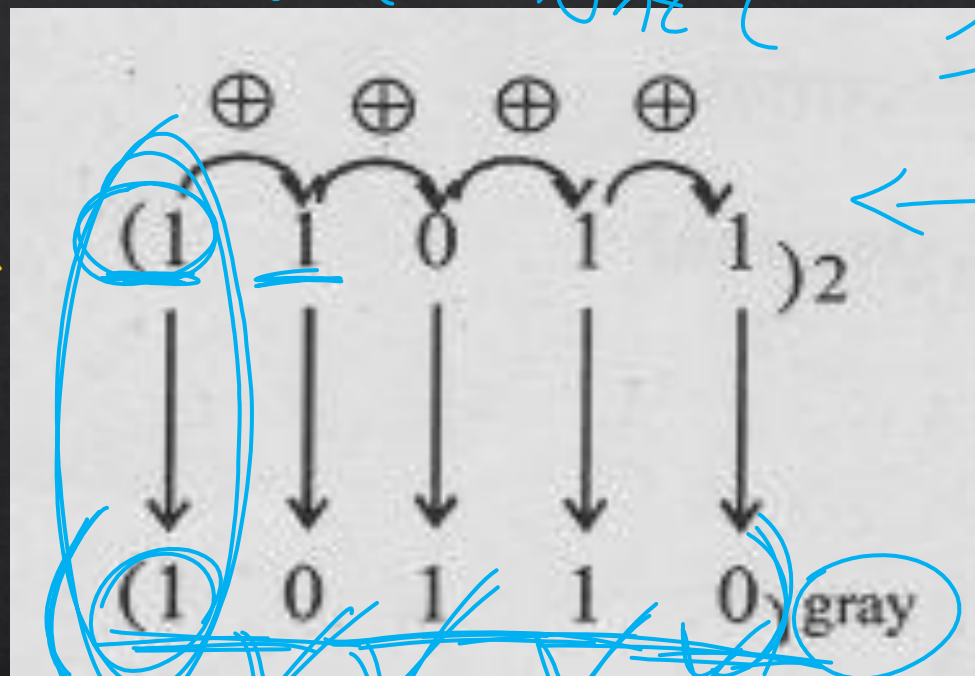
- Simplify the Boolean Function: $F(A, B, C) = \Sigma(1, 2, 3, 5, 7)$

And Draw Logic circuit of Simplified Expression

Convert given Binary code to gray code $(11011)_2$

Solution:

Left to right (→)



Binary

$$1 \oplus 1 = 0$$
$$1 \oplus 0 = 1$$
$$0 \oplus 1 = 1$$
$$1 \oplus 1 = 0$$

as it is

Convert Gray Code 1110 to Binary

- **Binary reconstruction rule:**

- First binary bit = first Gray bit
- Each next binary bit = previous binary bit $\oplus$ current Gray bit.

- Applying it:

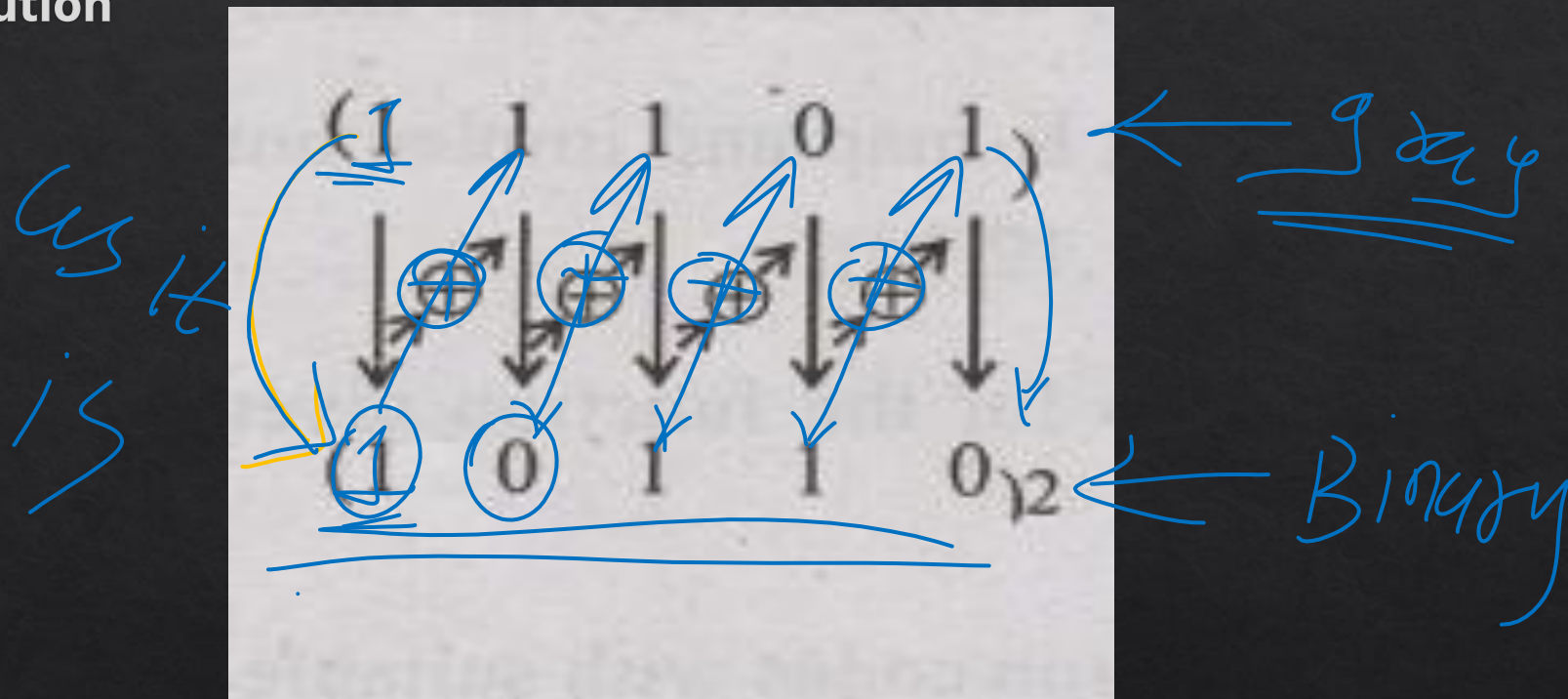
- $b_1 = 1$
- $b_2 = 1 \oplus 1$ (XOR your first binary bit (1) with the second Gray bit (1))
- $b_3 = 0 \oplus 1$ (XOR your second binary bit (0) with the third Gray bit (1))
- $b_4 = 1 \oplus 0$ (XOR your third binary bit (1) with the fourth Gray bit (0))

- So the binary value is **1011**.



Convert the gray code to binary $(1\ 1\ 1\ 0\ 1)_{\text{gray}}$

◇ Solution



Ex $(0\ 1\ 0\ 1\ 1)_{\text{gray}} \Rightarrow (?)_2 \Rightarrow (0\ 1\ 1\ 0\ 1)_2 \leftarrow \text{Ans}$